



Nil Kamleshbhai Makvana

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[in NilMakvana](#) [GitHub NilMakvana](#)



Education

Indraprastha Institute of Information Technology, Delhi

M.Tech in Computer Science and Engineering

2025 — Present

CGPA: 8.67

(till 2nd Semester)

Dharmsinh Desai University

B.Tech in Computer Engineering

2020 — 2024

CGPA: 8.59

Ashadeep Vidhyalaya

Gujarat Board, Standard 12, (PCM)

2019 — 2020

Percentage: 85.84%

Ashadeep Vidhyalaya

Gujarat Board, Standard 10

2017 — 2018

Percentage: 94%

Skills

Core Expertise:	Distributed Systems, System Programming, Backend Engineering
Languages:	C, C++, SQL, Python, Java, JavaScript, TypeScript
Frameworks:	gRPC, FastAPI, Django, OpenCV, LangChain, CrewAI
Tools & Technologies:	Linux, Git, GDB, Docker, GitHub, Postman
Databases:	PostgreSQL, MySQL, SQLite, MongoDB, Redis, Neo4j, ChromaDB
Coursework:	Graduate Algorithms, Operating Systems, OOP Design, Graduate Systems, Applied Cryptography, Computer Networks, Machine Learning

Work Experience

Jr. Software Development Engineer — Rejoicehub LLP

March 2024 — June 2025

Tech Stack: Python, FastAPI, PostgreSQL, SQLAlchemy, Alembic, REST APIs, JWT, Docker, CrewAI, LangChain, LLMs, Web Scraping, Machine Learning

- Built **Barbera**, a full-stack barbershop booking platform using **FastAPI** and **PostgreSQL**, delivering appointment scheduling, authentication, profile management, and service booking through **20+ REST APIs**.
- Developed **AI agents** using **CrewAI** and **LangChain**, automating travel planning and content generation workflows with LLM-based task orchestration.
- Engineered **web scraping and ETL pipelines** for automated data collection, processing, and structured knowledge extraction.
- Fine-tuned language models and built AI content-generation pipelines, reducing manual content creation effort while improving output quality.

Software Development Intern — Rejoicehub LLP

Dec 2023 — March 2024

Tech Stack: Python, FastAPI, PostgreSQL, SQLAlchemy, JWT, Passlib, bcrypt, Pandas, Regex, Streamlit, Plotly, TextBlob

- Engineered a **Social Media REST API** using **FastAPI**, **PostgreSQL**, and **SQLAlchemy**, implementing **20+ REST endpoints** and **8+ core features** including authentication, posts, stories, comments, and follow/unfollow functionality with OTP verification and **JWT access/refresh** token security.
- Developed a **WhatsApp Chat Analytics Platform** capable of processing **50,000+ messages**, generating **8+ analytical insights** including sentiment analysis, activity trends, emoji statistics, word-frequency analysis, and group/individual engagement metrics through interactive dashboards built with **Streamlit** and **Plotly**.

Projects

Distributed Peer-to-Peer (P2P) File Sharing System

April 2026 — June 2026

Tech Stack: C++, Socket Programming, Concurrency, Multi-threading, Performance Optimization, Networking, TCP/IP

- Architected a distributed **P2P File Sharing System** with **100+ concurrent peer connections**, leveraging POSIX multi-threaded socket programming and **Rarest-First chunk scheduling** to achieve a **45% increase in download throughput**.

Cellular Network Simulator with Multi-threaded Network Modeling

Oct 2025 — Nov 2025

Guide: Dr. Arani Bhattacharya

Team Size: 2

Tech Stack: C++, Design Patterns, Multithreading, Concurrency, System Programming, OOP

- Engineered a **Cellular Network Simulator** for **2G/3G/4G/5G** networks, leveraging object-oriented design, POSIX multithreaded **5G dual-band processing**, custom exceptions, templates, and system-level file handling to model communication infrastructure and optimize network resource allocation.

Mini Version Control System

May 2026 — June 2026

Tech Stack: C++, Hashing, File System, Zlib, Memory Optimization

- Implemented a **Git-inspired version control system** by implementing core commands including **init, add, commit, status, log, checkout, cat-file, hash-object, write-tree, and ls-tree**; leveraged **SHA-1 hashing**, compression, and content-addressable storage for blob, tree, and commit objects, achieving a **55% reduction in storage usage**.

Distributed LLM Communication Analysis

March 2026 — April 2026

Guide: Dr. Rinku Shah

Team Size: 3

Tech Stack: Python, Flask, gRPC, SimGrid, Pandas, Docker, NumPy, Matplotlib, REST APIs

- Developed a distributed **LLM training simulation framework** to benchmark **9 communication algorithms** across multi-node GPU clusters, conducting **576+ SimGrid-based experiments** through **REST/gRPC APIs** to identify scalable communication strategies.

Positions of Responsibility

- Teaching Assistant, DSA** — Supported **80+ students** through tutorials, labs, and programming mentorship.
- Deputy Head, Sports Council** — Led planning and execution of campus sports events, coordinating logistics and participation for **15+ inter-college competitions**.
- Mentor, Technical Expert** — Guided **50+ students** in **FastAPI, Django**, and **Git/GitHub**-based development projects.
- Volunteer, BAPS Swaminarayan Sanstha** — Led volunteer teams and organized community welfare programs.

Awards and Achievements

- Excellent TA Award - IIIT Delhi (Winter 2026)** - Recognized by the **Dean of Academic Affairs** for outstanding contribution and impact as a **Teaching Assistant**.
- AIR 2928 — GATE 2025 (CS & IT)** — Secured top **1.70%** position among **170,000+** participants.
- AIR 1992 — GATE 2025 (Data Science & AI)** — Ranked in top **3.50%** out of **57,000+** candidates.

Interests and Hobbies

- Teaching, Singing, Playing Instruments, Acting, Photography, Sports (Badminton), Reading Books

Declaration: The above information is correct to the best of my knowledge.

Name: Nil Kamleshbhai Makvana

Date: July 4, 2026